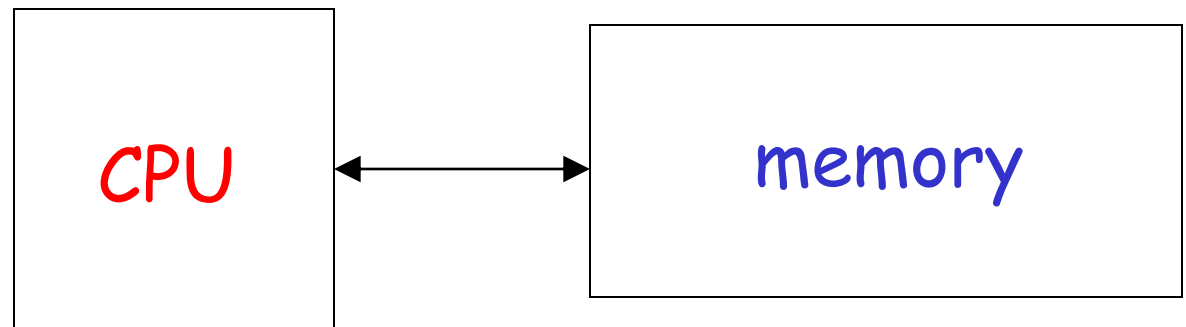
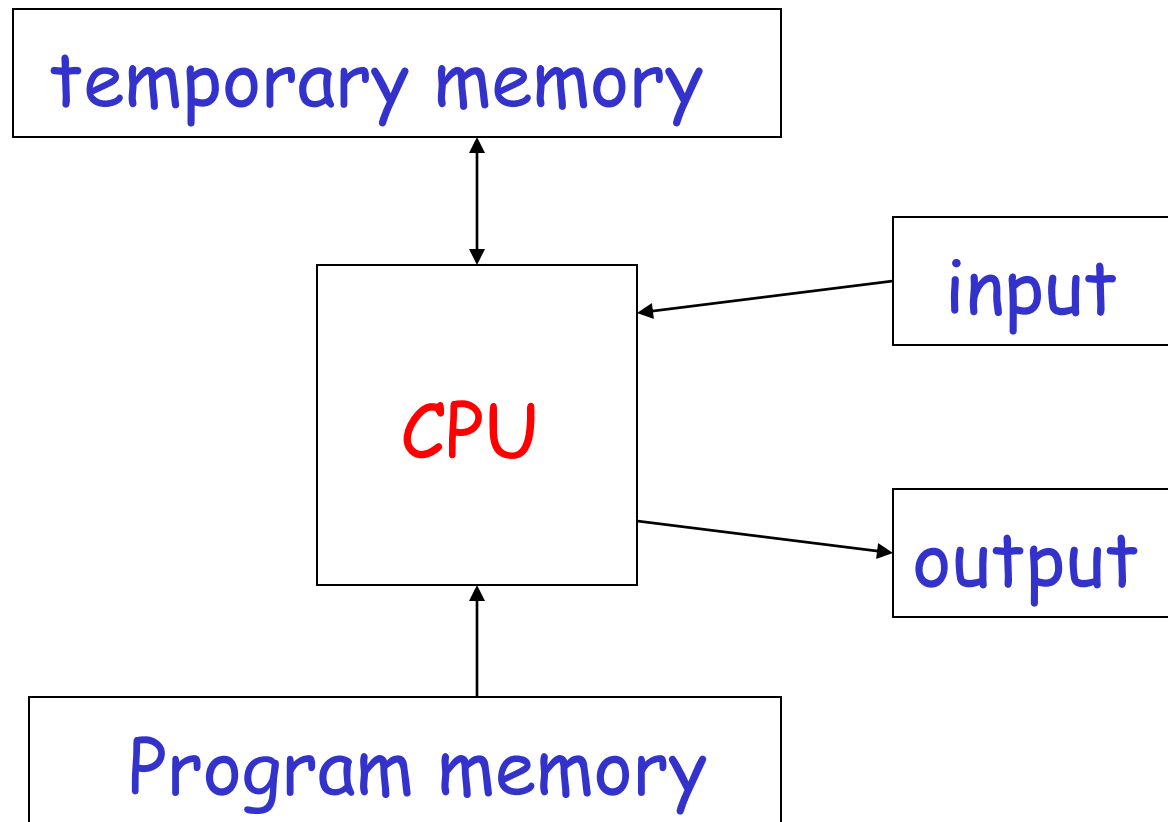


Models of Computation

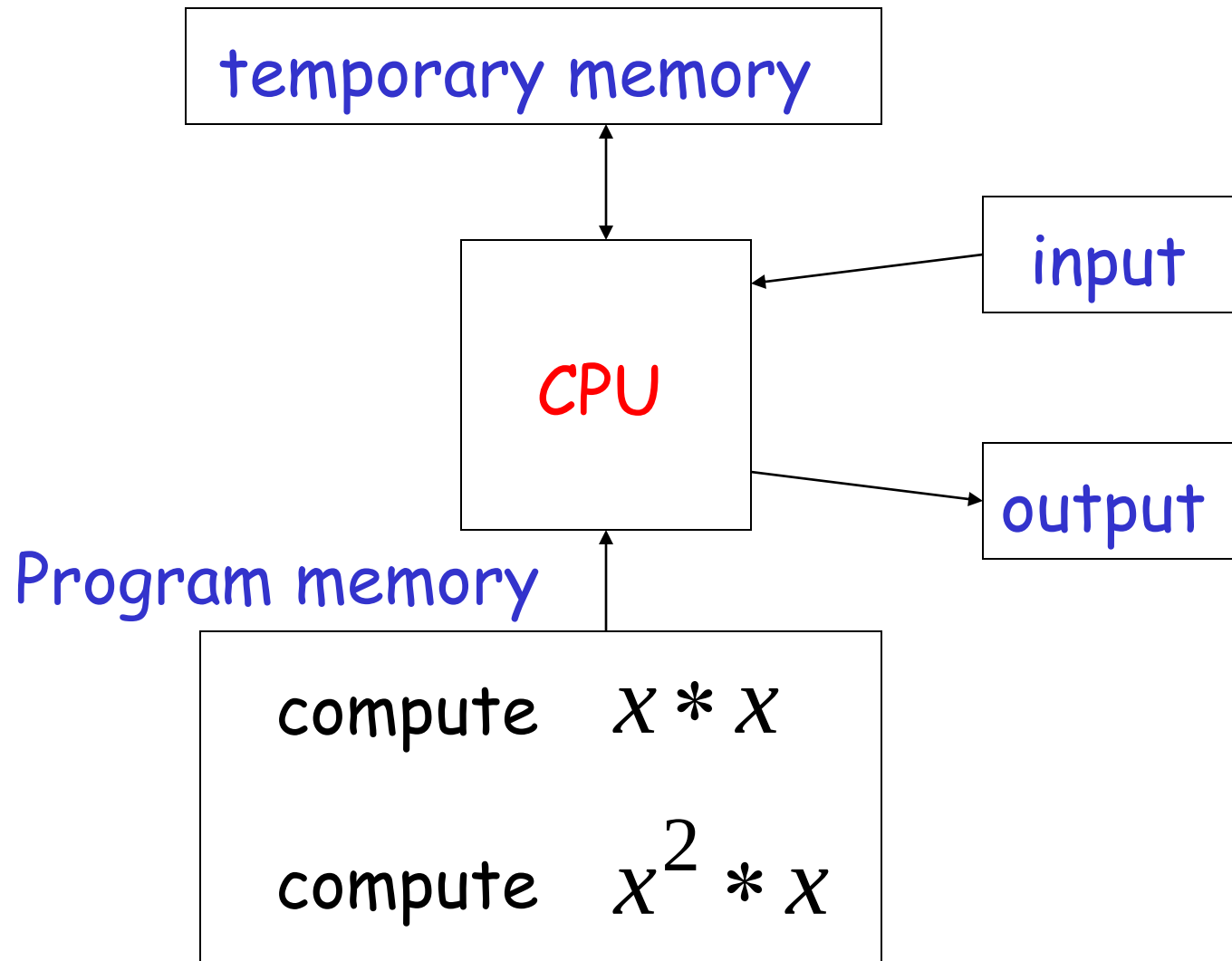
Outline of the course contents

Computation

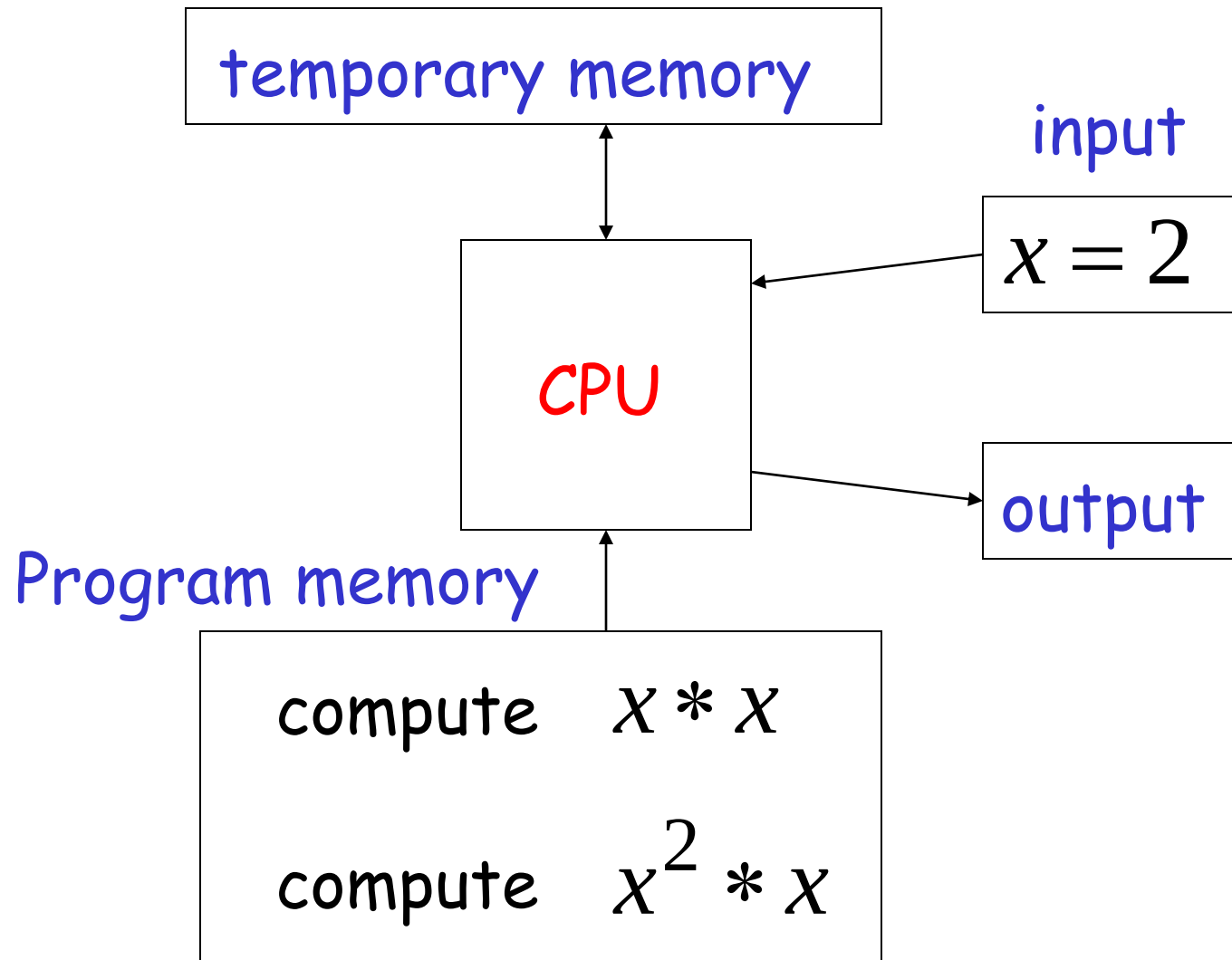




Example: $f(x) = x^3$



$$f(x) = x^3$$



$$f(x) = x^3$$

temporary memory

$$z = 2 * 2 = 4$$
$$f(x) = z * 2 = 8$$

input

$$x = 2$$

CPU

output

Program memory

compute $x * x$

compute $x^2 * x$

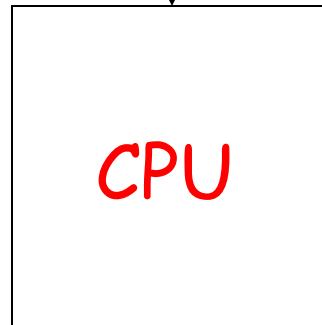
$$f(x) = x^3$$

temporary memory

$$z = 2 * 2 = 4$$
$$f(x) = z * 2 = 8$$

input

$$x = 2$$



$$f(x) = 8$$

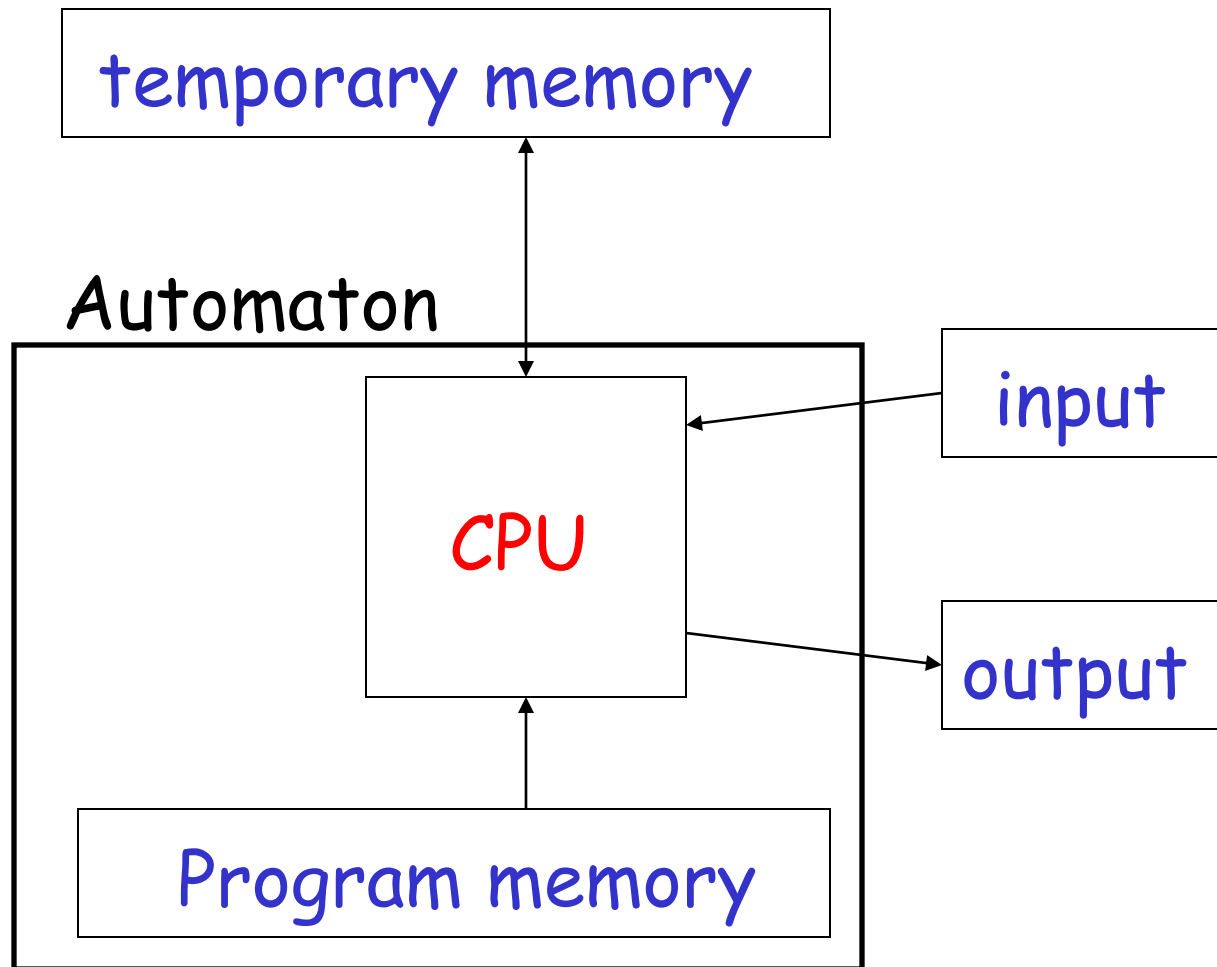
output

Program memory

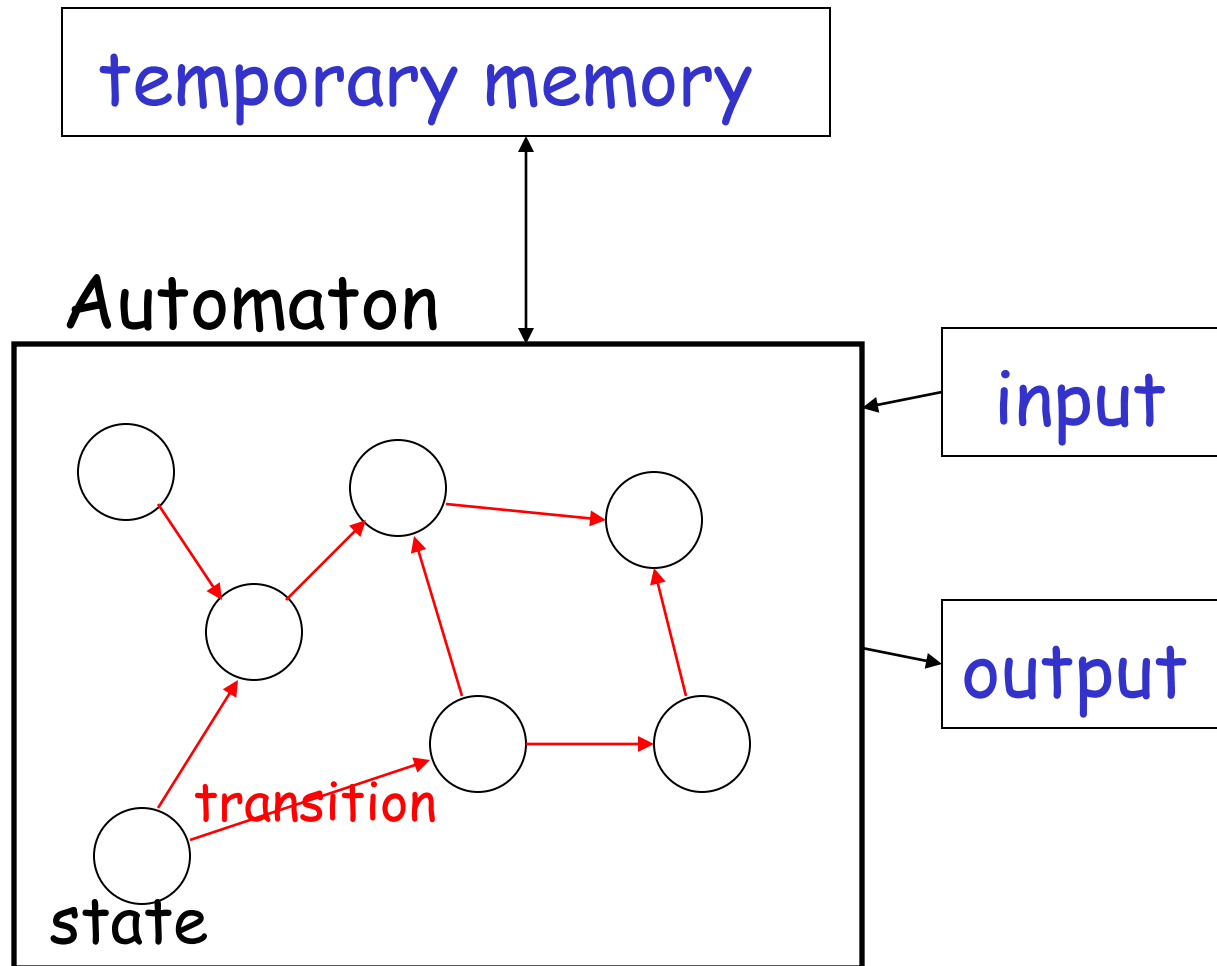
compute $x * x$

compute $x^2 * x$

Automaton



Automaton

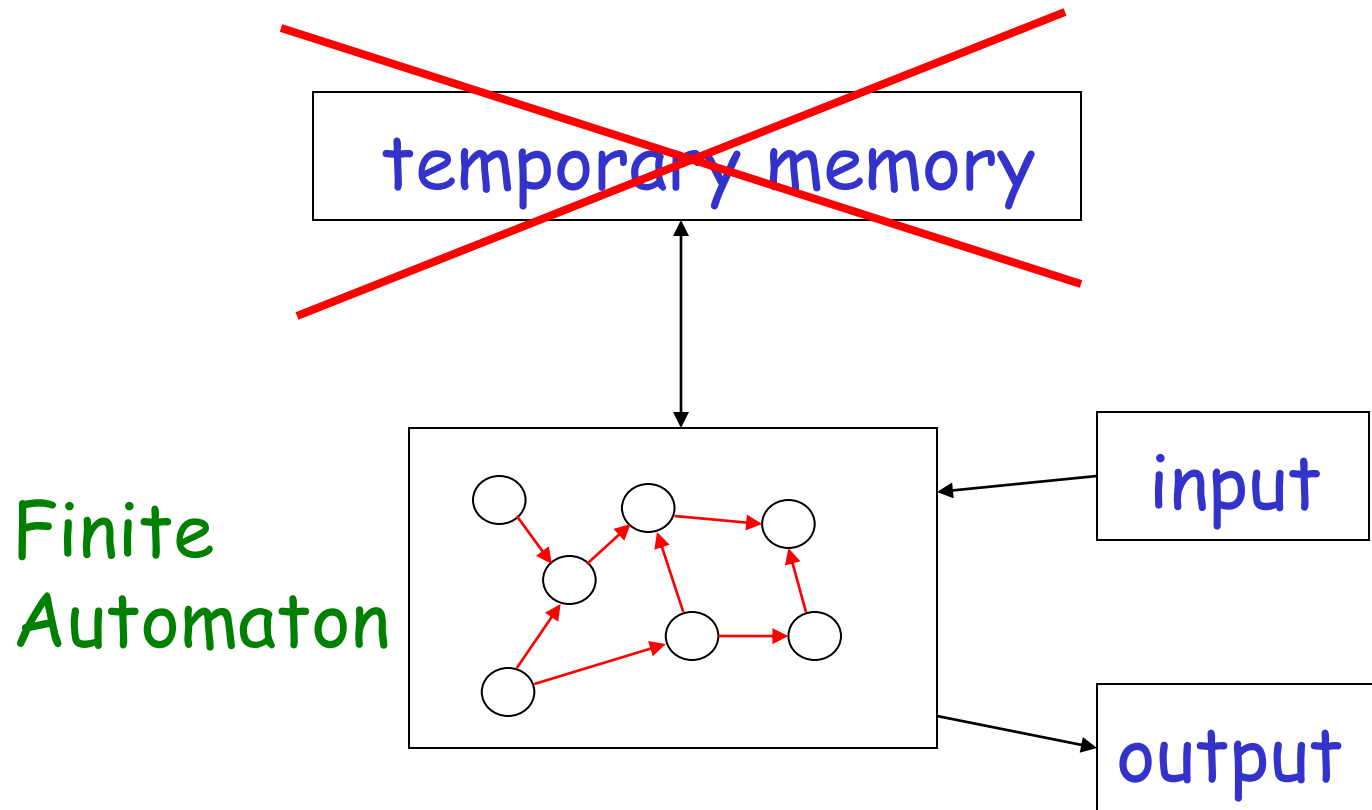


Different Kinds of Automata

Automata are distinguished by the temporary memory

- **Finite Automata:** no temporary memory
- **Pushdown Automata:** stack
- **Turing Machines:** random access memory

Finite Automaton



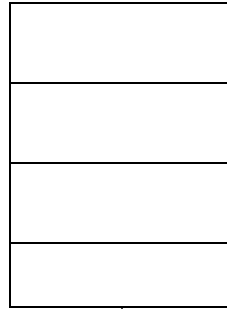
Example: Elevators, Vending Machines
(small computing power)

Pushdown Automaton

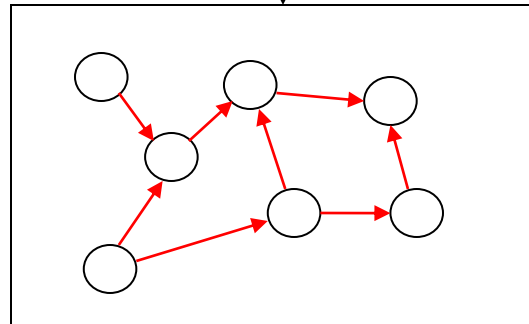
Temp.
memory

Stack

Push, Pop



Pushdown
Automaton



input

output

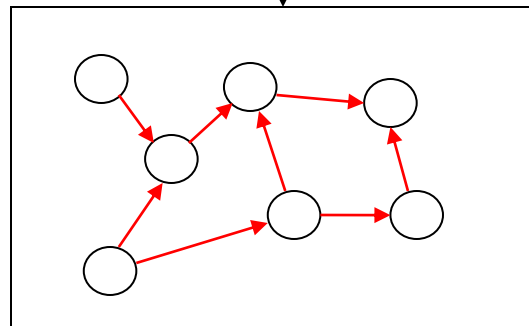
Example: Compilers for Programming Languages
(medium computing power)

Turing Machine

Temp.
memory

Random Access Memory

Turing
Machine



input

output

Examples: Any Algorithm

(highest computing power)

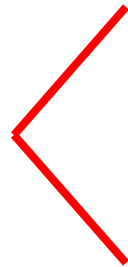
Power of Automata

Simple
problems

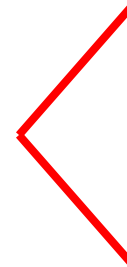
More complex
problems

Hardest
problems

Finite
Automata



Pushdown
Automata



Turing
Machine

Less power



More power

Solve more

computational problems

Turing Machine is the most powerful computational model known

Question: Are there computational problems that a Turing Machine cannot solve?

Answer: Yes (unsolvable problems)

Time Complexity of Computational Problems:

NP-complete problems

Believed to take exponential
time to be solved

P problems

Solved in polynomial time